بِسْمِ ٱللَّهِ ٱلرَّحْمَٰنِ ٱلرَّحِيمِ

In the name of Allah, Most Gracious, Most Merciful

CSE 4107
Structured Programming I

Topic: Introducing C & C fundamentals

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Programming Language

What is Language?

Language is a mode of communication that is used to share ideas, opinions with each other.

What is Programming Language?

A programming language is a computer language that is used by programmers (developers) to communicate with computers.
It is a set of instructions written in any specific language (C, C++, Java, Python) to perform a specific task.

Source: https://www.javatpoint.com/programming-language

Generations of Programming Language

- First Generation: Machine Language
  - Consist of binary code (0s and 1s) that directly corresponds to the internal instructions understood by a specific computer processor.
- Second Generation: Assembly Language
  - Uses symbolic instructions that are easier to understand and remember than binary code.

```
section     .text
global      _start                          ;must be declared for linker (ld)

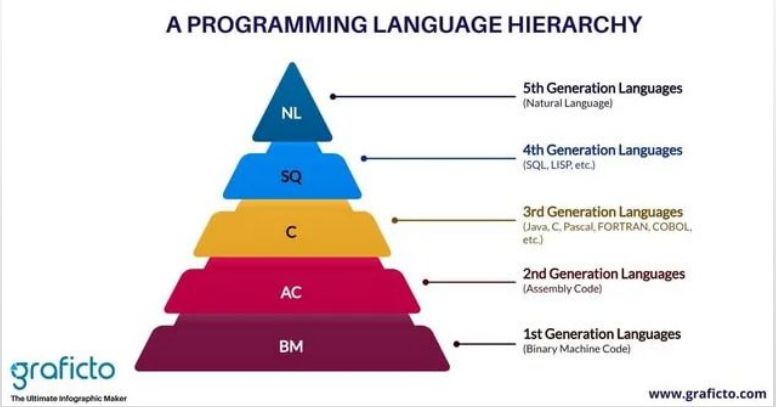
_start:                                     ;tell linker entry point

    mov     edx,len                         ;message length
    mov     ecx,msg                         ;message to write
    mov     ebx,1                           ;file descriptor (stdout)
    mov     eax,4                           ;system call number (sys_write)
    int     0x80                            ;call kernel

    mov     eax,1                           ;system call number (sys_exit)
    int     0x80                            ;call kernel

section     .data

msg     db  'Hello, world!',0xa             ;our dear string
len     equ $ - msg                         ;length of our dear string
```



Source: https://medium.com/@ajuatahcodingarena/generations-of-programming-languages-bed30d19ea8e

Source: https://medium.com/@jleveewhite/an-introduction-to-assembly-language-8144ce1dfb0e

Generations of Programming Language

- Third Generation: High-Level Languages
  - These languages use English-like keywords and syntax, making them much more readable and writable for humans.
- Fourth Generation: Very High-Level Languages
  - Even more user-friendly and domain-specific than third-generation languages. They often focus on specific tasks or problem domains, such as database management, web development, or scientific computing.
- Fifth Generation: Natural Language Programming
  - Aims to allow humans to communicate with computers using natural language like English. NLP is still in its early stages of development.

Source:
https://medium.com/@ajuatahcodingarena/generations-of-programming-languages-bed30d19ea8e

```sql
SELECT COUNT(CustomerID), Country
FROM Customers
GROUP BY Country
ORDER BY COUNT(CustomerID) DESC;
```

SQL: Structured Query Language

```cpp
#include <iostream>
using namespace std;

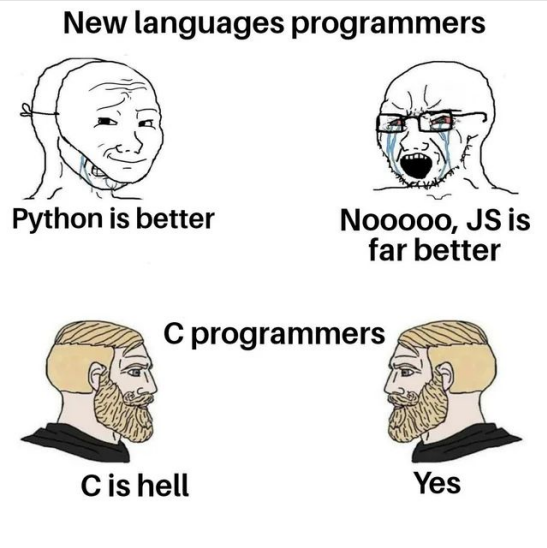
int main() {
  cout << "Hello World!";
  return 0;
}
```

C++

Introducing C

What is C?
A widely used programming language developed in the early 1970s at Bell Laboratories.

When someone says “I want a programming language in which I need only say what I wish done,” give him a lollipop.
(Alan Perlis)



Source: Reddit:r/ProgrammerHumor

History of C

Origin:

- C is a by-product of UNIX operating system, developed at Bell Lab by Ken Thompson, Dennis Ritchie and others.
- Ken Thompson wrote the original UNIX OS (written in assembly) for mini computer (DEC PDP-7) with 8K words main memory.
- For further development of UNIX OS a small high level language called B was created by Thompson.
- In 1970 Bell Lab got a new mini computer PDP-11 for the UNIX project. B was running in PDP-11
- When B was not well suited for PDP-11, Ritchie developed extended version of B called NB (New B)
- Later NB (New B) was renamed to C.
- In 1973 C became stable enough for rewriting UNIX.
- After creating C compilers for different machines, UNIX got running on those machines as well.

Standardization:

- During 1970s and 1977-1979 C continued evolving much. In 1978 first book on C was released.
- By 1980 C expanded beyond the UNIX OS.
- US standard for C began in 1983 under American National Standards Institute (ANSI). This version is referred to as C89 or C90
- Later the standard C99 came with some major differences from C89
- Today (3rd Sep 2024) the latest standard of C is C23.

Source: C Programming Modern Approach Book

Strengths and Weakness of C

Philosophy of C:

- **A low-level language**: Suitable for systems (OS) programming as it provides access to machine level concepts like bytes and addresses in memory. C programs are fast.
- **A small language:** Provides a more limited set of features than many other languages.
- **A permissive language**: Assumes that you know what you are doing, allowing you a wider degree of freedom.

Strength:

- **Efficiency:**C programs can run quickly in limited amounts of memory.
- **Portability:** C programs are portable from one type of computer to another. Thanks to the C compilers being small and easy to write.
- **Power:** Large collection of data types and operators.
- **Flexibility:** Flexible to use from System programming to all kinds of applications from embedded systems to commercial data processing.
- **Standard Library:** Contains hundreds of functions for different task.
- **Integration with UNIX:** Is powerful with combination of UNIX (like Linux)

Source: C Programming Modern Approach Book

Strengths and Weakness of C

Weakness:

- **Error-prone**: Programming mistakes that would be caught in many languages, can't be detected by C compilers. It's more like assembly language.
- **Difficult to understand:** Provides a more limited set of features than many other languages.

```
v,i,j,k,l,s,a[99];
main()
{
  for(scanf("%d",&s);*a-s;v=a[j*=v]-a[i],k=i<s,j+=(v=j<s&&
(!k&&!!printf(2+"\n\n%c"-(!l<<!j)," #Q"[l^v?(l^j)&1:2])&&
++l||a[i]<s&&v&&v-i+j&&v+i-j))&&!(l%=s),v||(i==j?a[i+=k]=0:
++a[i])>=s*k&&++a[--i])
    ;
}
```

Printing all solution to Eight Queen problem



Source: https://www.memelate.com/hijibiji-hizibizi_anis_zahid-hasan_meme-template/

- **Difficult to modify:** Lacks the feature of Class and Packaging for better maintainability.

Source: C Programming Modern Approach Book

C Fundamentals

Writing a Simple Program:

Follow the white board

Compiling and Linking:

3 steps to convert the program (written on the board) to machine executable form:

- **Preprocessing:** First the program (written as text) goes through a preprocessor that only process the commands beginning with # (known as directives). It is like an editor.
- **Compiling:** The modified version of the program goes to the compiler which translates it into the machine instructions (object code).
- **Linking:** A Linker combines the object code with additional code needed to yield a complete executable program.

Source: C Programming Modern Approach Book

C Fundamentals

General Form of a Simple Program:

Follow the white board

- Directives
- Functions
- Statements
- Printing Strings
- Blocks

Source: C Programming Modern Approach Book

C Fundamentals

Comments:

Follow the white board

- Single line comment
- Multiline comment
- Boxed comment
- Simplified boxed comment
- Uncaught error in multiline comment.
- Nested Comments

Source: C Programming Modern Approach Book

C Fundamentals

Variables and Assignments:

`Follow the white board`

- Types: int, float, char
- Rounding error in float
- Declarations: Same type, Different type
- Assignments: Order matters while assigning and declaring. Assignment at the time of declaration.
- Printing variable values: `printf` function, format specifier, digit control after decimal point.
- Initialization
- Printing Expressions

Source: C Programming Modern Approach Book

C Fundamentals

Reading Input:

Follow the white board

- Format String
- Reading input:`scanf` function, format specifier.

Source: C Programming Modern Approach Book

C Fundamentals

Defining Names for Constants:

Follow the white board

- Macro definition: `#define`
- Naming convention

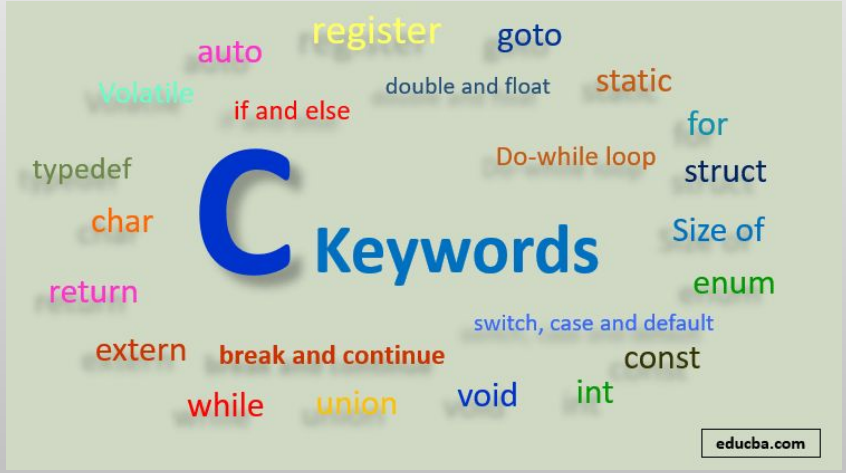
Source: C Programming Modern Approach Book

C Fundamentals

Identifiers:

Follow the white board

- Naming of variables, functions, macros
- May contain letters, digits and underscores
- Must begin with letter or underscore
- Case sensitivity
- No limits on the maximum length of identifiers (compiler dependence)
- An identifier cannot be a keyword



Source: C Programming Modern Approach Book

C Fundamentals

Layout of a C Program:

Follow the white board

- C program is a series of token.
- Identifiers, keywords, operators (like + - * /), punctuations (like comma, semicolon), string literals
- Indentations, Dividing statements in multiple line.
- Example:

```
printf  (  “Height: %d\n”  ,  height  )  ;
  1     2          3        4     5    6  7
```

7 Tokens. Token 1 and 5 are identifiers, token 3 string literal, token 2,4,6,7 are punctuation tokens

Source: C Programming Modern Approach Book


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A


THANK
YOU